PROJECT DEVELOPMENT & MANAGEMENT PLAN

Wildfire Risk Prediction Using Climate & Environmental Data

Master of Data Science · Final Year Project

Student Name	Maragani Gireesh
Programme	Master of Data Science
Project Title	Wildfire Risk Prediction Using Climate & Environmental Data
Dataset	1.88M US Wildfires — Kaggle (rtatman/188-million-us-wildfires)
Start Date	26 May 2026
End Date / Deadline	10July 2026
Total Duration	7 Weeks
Supervisor	Andy Tai Hock Lin

Project Links

GitHub	https://github.com/Gireesh462/Data-Science-Project
Google Drive	https://drive.google.com/drive/folders/1qNq5ZLyEogtSL4f5DMWdhrM3tz9bf8xu

Project Overview

Wildfires pose a serious threat to lives, ecosystems, and infrastructure across the United States. This project applies machine learning to predict whether a wildfire event is High Risk or Low Risk, using 1.88 million historical wildfire records spanning 1992 to 2015.

Four classification models are built and evaluated — Ridge Classifier, Random Forest, XGBoost, and MLP Neural Network. Class imbalance is handled using Balanced Bagging Classifier, and XGBoost is fine-tuned with Optuna hyperparameter optimisation. ELI5 explainability is applied to identify the environmental features that most strongly influence wildfire risk.

The project is evaluated using Accuracy, F1-Score, ROC-AUC, and 5-Fold Stratified Cross-Validation to ensure robust and reliable results. All work is implemented in Python on Google Colab, with code maintained on GitHub and outputs stored on Google Drive.

1. Research Questions

Main Research Question:

"How accurately can ensemble models and neural networks classify wildfire risk using historical climate and environmental data?"

Project Objectives:

OBJ – 1: Compare ensemble models and neural network to identify the best-performing model for wildfire risk prediction by ROC-AUC and F1-score.
OBJ – 2: Identify the strongest environmental predictors of high-risk wildfire events using ELI5 explainability analysis

2. Project Steps

- Download the 1.88M US Wildfire dataset directly from Kaggle using the kagglehub API.
- Clean and preprocess data — missing values, encoding, duplicates.
- Engineer features: SEASON, RISK_BINARY, FIRE_SIZE_LOG, REGION.
- Apply Balanced Bagging Classifier to handle the 90/10 class imbalance.
- Produce 5 EDA charts and 1 interactive geospatial Folium heatmap.
- Train 4 models: Ridge Classifier (baseline), Random Forest, XGBoost, MLP Neural Network.
- Fine-tune XGBoost using Optuna to maximise ROC-AUC.
- Evaluate with Accuracy, F1, ROC-AUC, PR Curve, and 5-Fold Stratified Cross-Validation.
- Apply ELI5 to identify top environmental predictors of wildfire risk.
- Push all code to GitHub and save all outputs to Google Drive.
- Submit full thesis by 10 July 2026.

3. Detailed Week-by-Week Plan

Week	Dates	Phase	Tasks	Deliverable
1	26 May – 1 Jun	Project Setup	Configure Kaggle API, install libraries. Download dataset via kagglehub. Auto-detect SQLite table. Inspect all columns. Create GitHub repo and Google Drive project folder.	Environment ready + dataset downloaded
2	2 Jun – 5 Jun	Data Cleaning	Handle missing values. Drop rows missing FIRE_SIZE_CLASS, LATITUDE, LONGITUDE, STATE, STAT_CAUSE_DESCR. Fill OWNER_DESCR nulls. Remove duplicates. Validate clean dataset. Push notebook to GitHub.	Clean dataset ready
3	6 Jun – 10 Jun	Feature Eng. + EDA	Create SEASON, RISK_BINARY, FIRE_SIZE_LOG, REGION. Apply Balanced Bagging. Scale with StandardScaler. Produce 5 EDA plots + Folium heatmap. Save all plots to Google Drive.	Feature matrix + 6 plots saved

Week	Dates	Phase	Tasks	Deliverable
4	11 Jun – 22 Jun	Model Training	Train Model 1: Ridge Classifier (baseline). Train Model 2: Random Forest. Train Model 3: XGBoost. Train Model 4: MLP Neural Network. Record Accuracy, F1, ROC-AUC for all 4 models.	4 trained models + scores
5	23 Jun – 29 Jun	Optuna Tuning + Evaluation	Fine-tune XGBoost with Optuna. Run 5-fold stratified CV on all models. Plot confusion matrix, ROC curve, PR curve. Apply ELI5 explainability on best model. Print all classification reports. Save outputs to Google Drive.	Tuned XGBoost + eval plots
6	30 Jun – 6 Jul	Report Ch. 1–3	Save all models with joblib. Write Chapter 1 (Introduction). Write Chapter 2 (Literature Review). Write Chapter 3 (Dataset & EDA). Save thesis draft to Google Drive.	Ch. 1–3 drafted
7	7 Jul – 10 Jul	Report Completion	Write Chapters 4–7 (Methodology, Results, Discussion, Conclusion). Proofread. Final push to GitHub and Google Drive. Submit to supervisor by 15 July 2026.	FINAL SUBMISSION ✓

4. Techniques We Are Using & Why

No.	Technique	Why We Are Using This
1	Ridge Classifier	Baseline model with L2 regularisation. Simple and fast — all other models must beat it.
2	Random Forest	Handles noisy wildfire data well. Reduces overfitting. Strong on tabular structured data.
3	XGBoost	Best performer on structured tabular data. Fine-tuned with Optuna for maximum accuracy.
4	MLP Neural Network	Captures complex non-linear patterns between climate features. Represents deep learning comparison.
5	Balanced Bagging	Fixes 90/10 class imbalance by balancing each bootstrap sample automatically.
6	Optuna	Intelligently tunes XGBoost parameters — faster and smarter than Grid Search.
7	ELI5	Explains model predictions in plain English — shows which features drive High Risk predictions.

No.	Technique	Why We Are Using This
8	5-Fold Stratified CV	Tests the model 5 times on different data splits to confirm results are stable.
9	Folium + Matplotlib	Folium for interactive geographic heatmap. Matplotlib for all EDA and evaluation charts.

5. References

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